Integration

ANSWER KEY

Integration by substitution

1 Find $\int 2x(x^2+1)^4\,dx$ using the substitution $u = x^2+1$.
$du = 2x\,dx$. Integral becomes $\int u^4\,du = \frac{u^5}{5} + C = \frac{(x^2+1)^5}{5} + C$.

2 Find $\int x\sqrt{x^2+9}\,dx$ using the substitution $u = x^2+9$.
$du = 2x\,dx$, so $x\,dx = \frac{du}{2}$. Integral becomes $\int \sqrt{u}\cdot\frac{du}{2} = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = \frac{1}{3}(x^2+9)^{3/2} + C$.

3 Find $\int (2x+1)(x^2+x)^3\,dx$ using the substitution $u = x^2+x$.
$du = (2x+1)\,dx$. Integral becomes $\int u^3\,du = \frac{u^4}{4} + C = \frac{(x^2+x)^4}{4} + C$.

4 Find $\int 3x^2(x^3-2)^4\,dx$ using the substitution $u = x^3-2$.
$du = 3x^2\,dx$. Integral becomes $\int u^4\,du = \frac{u^5}{5} + C = \frac{(x^3-2)^5}{5} + C$.

Integration by parts

5 Find $\int xe^x\,dx$ using integration by parts.
Let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C$.

6 Find $\int x\cos x\,dx$ using integration by parts.
Let $u = x$, $dv = \cos x\,dx$, so $du = dx$, $v = \sin x$. $\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$.

7 Find $\int \ln x\,dx$ using integration by parts with $u = \ln x$ and $dv = dx$.
$du = \frac{1}{x}dx$, $v = x$. $\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - \int 1\,dx = x\ln x - x + C$.

8 Find $\int x^2e^x\,dx$ using integration by parts twice.
Let $u = x^2$, $dv = e^x\,dx$: $\int x^2e^x\,dx = x^2e^x - \int 2xe^x\,dx$. From an earlier result, $\int xe^x\,dx = e^x(x-1) + C$, so $\int 2xe^x\,dx = 2e^x(x-1) + C$. Result: $x^2e^x - 2e^x(x-1) + C = e^x(x^2-2x+2) + C$.

Integration using partial fractions

9 Find $\int \frac{5x+7}{(x+1)(x+2)}\,dx$, first decomposing into partial fractions.
$\frac{5x+7}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$: $5x+7 = A(x+2) + B(x+1)$. At $x = -1$: $A = 2$. At $x = -2$: $B = 3$. Integral: $2\ln|x+1| + 3\ln|x+2| + C$.

10 Find $\int \frac{1}{x^2-4}\,dx$ using partial fractions, factoring $x^2-4 = (x-2)(x+2)$.
$\frac{1}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}$: $1 = A(x+2) + B(x-2)$. At $x = 2$: $A = \frac{1}{4}$. At $x = -2$: $B = -\frac{1}{4}$. Integral: $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$.

11 Find $\int \frac{3x-1}{x(x-3)}\,dx$, first decomposing into partial fractions.
$\frac{3x-1}{x(x-3)} = \frac{A}{x} + \frac{B}{x-3}$: $3x-1 = A(x-3) + Bx$. At $x=0$: $A = \frac{1}{3}$. At $x=3$: $B = \frac{8}{3}$. Integral: $\frac{1}{3}\ln|x| + \frac{8}{3}\ln|x-3| + C$.

12 Find $\int \frac{2x+3}{(x-1)(x+4)}\,dx$, first decomposing into partial fractions.
$\frac{2x+3}{(x-1)(x+4)} = \frac{A}{x-1} + \frac{B}{x+4}$: $2x+3 = A(x+4) + B(x-1)$. At $x=1$: $A=1$. At $x=-4$: $B=1$. Integral: $\ln|x-1| + \ln|x+4| + C$.

Integrating trigonometric, exponential and logarithmic functions

13 Find $\int \sin(3x)\,dx$ and $\int e^{2x}\,dx$.
$\int \sin(3x)\,dx = -\frac{1}{3}\cos(3x) + C$. $\int e^{2x}\,dx = \frac{1}{2}e^{2x} + C$.

14 Find $\int \frac{4x}{x^2+1}\,dx$.
Since $\frac{d}{dx}[\ln(x^2+1)] = \frac{2x}{x^2+1}$, and $\frac{4x}{x^2+1} = 2 \cdot \frac{2x}{x^2+1}$: $\int \frac{4x}{x^2+1}\,dx = 2\ln(x^2+1) + C$.

15 Find $\int \tan x\,dx$, using the substitution $u = \cos x$.
$\tan x = \frac{\sin x}{\cos x}$. With $u = \cos x$, $du = -\sin x\,dx$:
$\int \frac{\sin x}{\cos x}\,dx = -\int \frac{du}{u} = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

16 Find $\int \frac{1}{2x+3}\,dx$.
$\int \frac{1}{2x+3}\,dx = \frac{1}{2}\ln|2x+3| + C$.

Definite integrals using substitution

17 Evaluate $\int_0^2 x(x^2+1)^3\,dx$ using the substitution $u = x^2+1$.
When $x=0$, $u=1$; when $x=2$, $u=5$; $x\,dx = \frac{du}{2}$. $\int_1^5 u^3 \cdot \frac{du}{2} = \frac{1}{2}\left[\frac{u^4}{4}\right]_1^5 = \frac{1}{8}(625-1) = \frac{624}{8} = 78$.

18 Evaluate $\int_0^{\pi/2} \sin x \cos^3 x\,dx$ using the substitution $u = \cos x$.
$du = -\sin x\,dx$. When $x=0$, $u=1$; when $x = \frac{\pi}{2}$, $u=0$. $\int_1^0 u^3(-du) = \int_0^1 u^3\,du = \left[\frac{u^4}{4}\right]_0^1 = \frac{1}{4}$.

19 Evaluate $\int_1^e \frac{\ln x}{x}\,dx$ using the substitution $u = \ln x$.
$du = \frac{1}{x}dx$. When $x=1$, $u=0$; when $x=e$, $u=1$. $\int_0^1 u\,du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}$.

Volumes of revolution: the disk method

20 The region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis. Find the volume generated, using $V = \pi\int y^2\,dx$.
$V = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = 8\pi$.

21 The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis. Find the volume.
$V = \pi\int_0^2 x^4\,dx = \pi\left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5}$.

22 The region under $y = e^x$ on $[0, 1]$ is revolved about the x-axis. Find the volume, using $V = \pi\int_0^1 e^{2x}\,dx$.
$V = \pi\left[\frac{e^{2x}}{2}\right]_0^1 = \pi\left(\frac{e^2}{2} - \frac{1}{2}\right) = \frac{\pi}{2}(e^2 - 1)$.

Volumes of revolution: washer and shell methods

23 The region between $y = x^2$ and $y = 9$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.
Bounds: $x^2 = 9 \Rightarrow x = \pm 3$. $V = \pi\int_{-3}^{3}[9^2 - (x^2)^2]dx = \pi\int_{-3}^{3}(81 - x^4)\,dx = \pi\left[81x - \frac{x^5}{5}\right]_{-3}^{3} = \frac{1944\pi}{5}$.

24 The region under $y = x^3$ on $[0, 2]$ is revolved about the y-axis. Use the shell method, $V = 2\pi\int_0^2 x \cdot x^3\,dx$, to find the volume.
$V = 2\pi\int_0^2 x^4\,dx = 2\pi\left[\frac{x^5}{5}\right]_0^2 = 2\pi\left(\frac{32}{5}\right) = \frac{64\pi}{5}$.

A well-designed word problem: designing a vase

25 This question has three parts. A vase is formed by rotating the curve $y = \sqrt{x}$, for $0 \le x \le 9$ (cm), about the y-axis, where x is horizontal distance from the axis and y is height. (a) Express x^2 in terms of y, and evaluate $V = \pi\int_0^3 x^2\,dy$ to find the volume of the vase. (b) A cylindrical base of radius 3 cm and height 2 cm is attached underneath. Find the total volume, to two decimal places (using $\pi \approx 3.14159$). (c) Show that the volume of water filled to height h (ignoring the base) is $V(h) = \frac{\pi h^5}{5}$, and find the height h at which the water volume equals half the vase's volume from part (a), to two decimal places.

(a) Since $y = \sqrt{x}$, $x = y^2$, so $x^2 = y^4$, and y ranges from 0 to 3 (as $x = 9 \Rightarrow y = 3$).
$V = \pi\int_0^3 y^4\,dy = \pi\left[\frac{y^5}{5}\right]_0^3 = \frac{243\pi}{5} = 48.6\pi$. (b) Cylinder volume: $\pi(3)^2(2) = 18\pi$. Total: $48.6\pi + 18\pi = 66.6\pi \approx 66.6(3.14159) \approx 209.23$ cm^3. (c) $V(h) = \pi\int_0^h y^4\,dy = \frac{\pi h^5}{5}$. Setting this equal to half of 48.6π, i.e. 24.3π: $\frac{h^5}{5} = 24.3 \Rightarrow h^5 = 121.5 \Rightarrow h = 121.5^{1/5} \approx 2.61$ cm.